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Review Article

Automated latent fingerprint identification system: A review

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ABSTRACT

Latent fingerprints are considered as one of the important evidences obtained from the site of crime. The process of developing, acquiring, processing and matching of latent fingerprints is different from the inked or live-scan fingerprints. Automated identification of latent fingerprints is still in its nascent phase when compared with the Automatic Fingerprint Identification System (AFIS) used by the police department. This paper provides an extensive review of the work done by eminent researchers in the development of an automated latent fingerprint identification system.

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1. Introduction

In today's modern world, biometrics has gained immense importance and acceptance across the globe for accurately identifying and authenticating an individual. The word 'biometric' represents biological measurements, which include physiological and behavioral characteristics such as fingerprint, hand geometry, face, voice, iris, gait, keystroke dynamics, signature, as shown in Fig. 1 [1].

Over the years, there has been enormous advancement in biometric technologies. As a result, it is widely integrated with numerous applications making operations, transactions and everyday life both safer and more convenient. Some areas currently deploying biometric technology are airport and border security; time and attendance systems; forensics and law enforcement; commercial sector; mobile biometrics; physical access control and surveillance.

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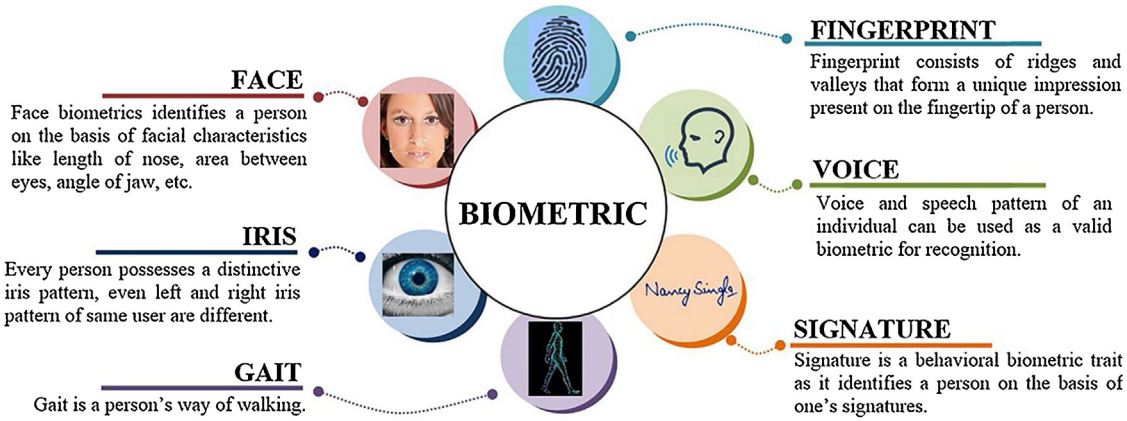


Fig. 1. Biometric characteristics.

A fingerprint consists of the unique pattern possessed by every individual [2]. It is the most prominent and widely accepted biometric trait used for establishing and verifying the identity of an individual due to its uniqueness, permanence, high indispensability, high universality, high collectability, good storability and high performance [3]. For centuries, fingerprints have been used in investigations for identifying and convicting criminals. In earlier times, fingerprints of the criminals or suspects were collected using ink, slab, roller and preserved on a fingerprint 10-print card. The number of criminal cases increases with the rising crime over time and hence, it becomes difficult to maintain and search these 10-print cards manually. With the evolution of technology, the Automated Fingerprint Identification System (AFIS) is developed. AFIS refers to a computer system developed to aid the process of maintaining, searching and matching fingerprints by digitizing the 10-print cards in computerized databases. Thereafter, AFIS systems were implemented on a large scale to improve and expedite the process of identification. Table 1 shows the comparison of before and after the development of AFIS [4].

Biometrics have been used in forensic applications for a long time. Finger impressions inadvertently left at the crime scenes by the culprits are known as the latent fingerprints and have been considered as prime evidence during the criminal investigation. Latent fingerprints on the surface are the results of the deposition of various residues and natural secretions from the skin [5]. Some samples of the latent fingerprints of different quality (good, bad and ugly) are shown in Fig. 2. These finger impressions are often not directly visible.

Fig. 3 shows the various stages involved in preparing the fingerprints, which will be available as input to AFIS. Firstly, at a crime scene, potential objects which might be in contact with the fingers of the suspects are visually inspected using bright light, forensic light or laser light to locate and visualize the latent fingerprints. Further, investigators use different methods including specialized dusting powders, chemical or optical methods for developing these located fingerprints depending upon the surface on which it has been found. After developing the latent fingerprints, these are first photographed. Fingerprints developed using dusting powders are further lifted using specialized tapes. These lifted prints are then affixed to the fingerprint card to preserve them. After that, these prints are digitized by scanning the fingerprint card. Photographed fingerprints or the prints obtained by digitizing the fingerprint card can be further enhanced to get a clearer image and fed as an input to the AFIS.

Automatic Fingerprint Identification System consists of two main types of searches, namely ten-print search (rolled and slap impressions) and a latent search. Due to technological advancements, the ten-print identification process of AFIS is fully automated with improved accuracy and throughput, leading to wide deployment in law enforcement agencies. According to the periodic evaluations performed by the National Institute of Standards and Technology (NIST), the False Negative Identification Rate (FNIR) is reported as 1.9% and False Positive Identification Rate (FPIR) is reported as 0.1% for matching rolled and slap prints [7]. In contrast to this, according to the evaluation performed by the NIST for latent fingerprints (ELFT-EFS), Rank-1 Identification Rate is 67.2% for searching 1114 latent prints against 100,000 reference images [8]. Moreover, in comparison with ten-print identification, the involvement of a fingerprint expert is required for extracting and matching the features from the latent fingerprint.

This paper presents a comprehensive review of the various stages involved in developing an automated system for latent fingerprint identification. The structure of the paper is organized as follows. Section 2 discusses different methods for developing latent fingerprints. Section 3 mentions the latent fingerprint identification system explaining both the manual process and the semi-automated process for identifying the latent fingerprints. Section 4 presents the literature review conducted on various stages of the automated latent fingerprint identification system.

Table 1
Comparison of before and after the development of AFIS.

	Before AFIS	After AFIS
Storage	Fingerprint 10-print cards are used to preserve the fingerprints	Fingerprint samples are digitally stored on RAID
Acquiring	Fingerprints on 10-print cards are acquired using ink, slab and roller	Fingerprints are acquired through live scanning using fingerprint sensors
Feature marking	Fingerprints features are classified and marked manually	Fingerprint features (minutiae points) are automatically extracted using the coder
Analysis	Fingerprints are searched and analyzed manually	Fingerprints are searched and analyzed automatically
Searching databases	Possible to search only local files	Possible to search local, state as well as national databases
Response time	It took some days or hours to generate a response	It takes a few minutes to generate a response

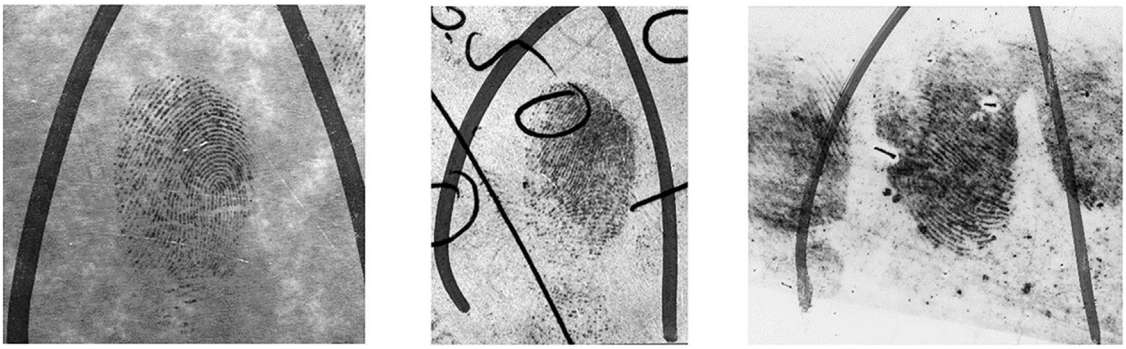


Fig. 2. Sample of latent fingerprints from NIST SD27 database [6].

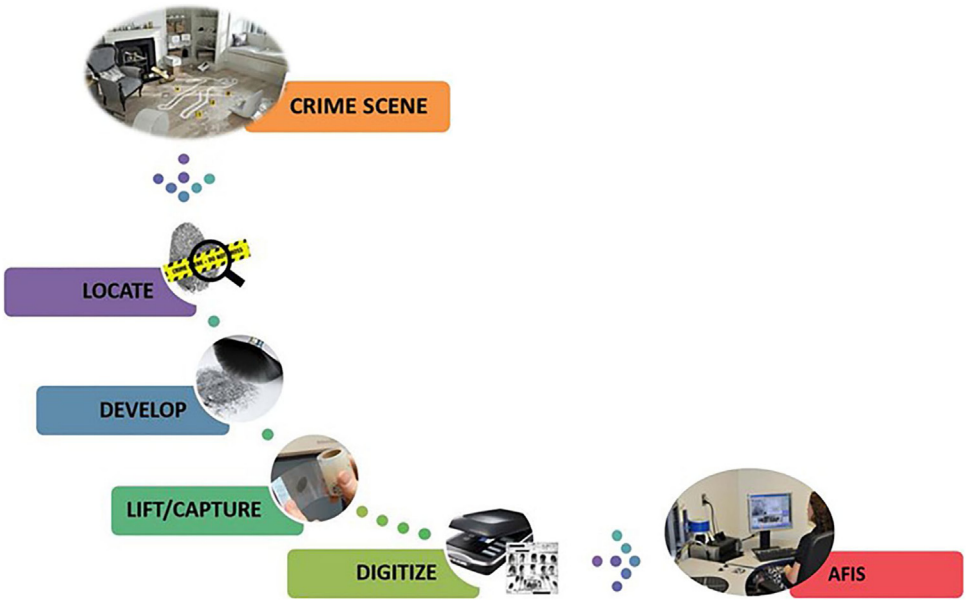


Fig. 3. Pipeline of various stages involved in preparing latent fingerprints from crime scene to AFIS.

Further, Section 5 enumerates the databases available for latent fingerprints. Finally, in Section 6, conclusions are drawn.

2. Methods of developing latent fingerprints

The purpose of developing latent fingerprints found at the crime scene is to make them visible to the naked eye. Numerous methods are available for developing latent fingerprints. However, the selection of the available methods highly depends on the nature of the surface from which the print needs to be developed [9]. Generally, there are two kinds of surfaces, namely non-porous and porous. The evolution of the commonly used methods for developing latent fingerprints for non-porous and porous surfaces is shown in Fig. 4 [10].

2.1. Methods for non-porous surfaces

Non-porous surfaces are the surfaces that are resistant to the passage of air or liquid through it, such as glass, plastics, metals, varnished woods. As a result, the finger impression resides on the non-porous surface itself. Methods for latent fingerprint development are widely categorized as powder method, chemical method and optical method. Different types of powders such as white, silver/grey, bi-chromatic, black, fluorescent, magnetic are available for visualizing the latent fingerprints [11]. Vacuum Metal Deposition (VMD) [12], Cyanoacrylate fuming and Multi-metal deposition (MMD) [9] are some popular chemical methods for developing latent fingerprints. There are also some less destructive contact-less acquisition methods such as a Reflected Ultra Violet Imaging System (RUVIS) and EVISCAN system for developing the latent fingerprints by exploiting the optical properties [9].

2.2. Methods for porous surfaces

Porous surfaces are surfaces on which the latent print is absorbed into the material. The dusting technique using powders is not much suitable and effective on porous surfaces such as cloth, wood, paper. Different types of specialized chemical methods such as silver nitrate reagent, ninhydrin chemical treatment, physical developer reagent, multi-metal deposition (MMD), DFO (1,8-Diazafluoren-9-One) and Oil Red O (ORO) [13] are often used by the investigators to locate and develop the latent prints on porous surfaces.

After developing the latent prints using a suitable method, these prints are lifted or captured for analysis and identification process.

3. Latent fingerprint identification system

In earlier times, the fingerprint experts manually performed the entire process of analyzing and matching the latent prints. However, over the years, some manual processes got automated,

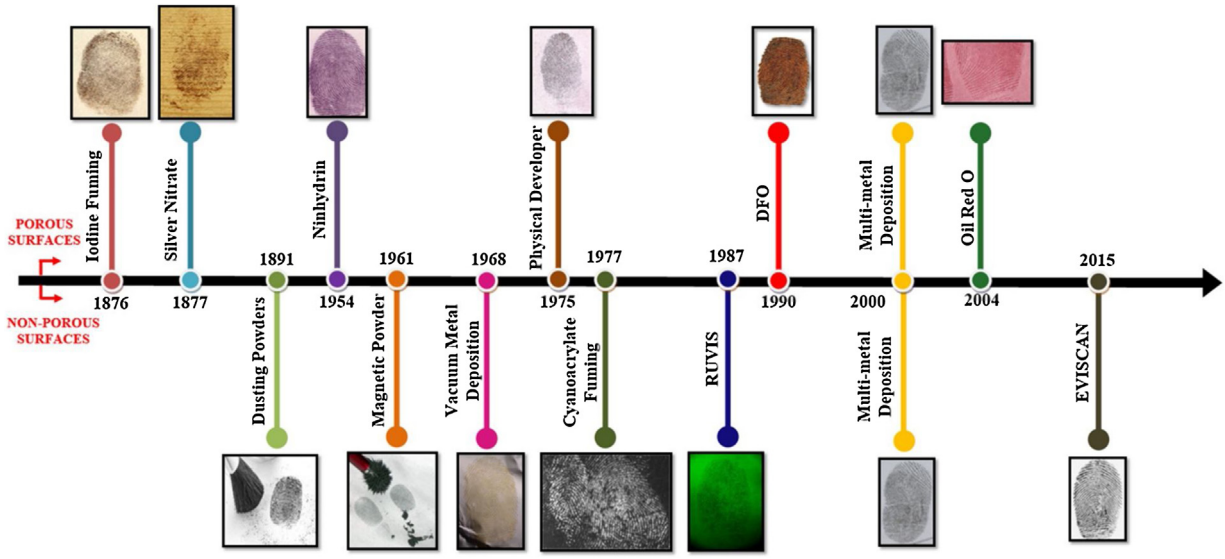


Fig. 4. Evolution of commonly used latent fingerprint development methods [10].

leading to the semi-automated latent fingerprint identification system. This section provides insight of the manual procedure and the semi-automated procedure for analyzing the latent fingerprints.

3.1. Latent fingerprint manual analysis

Latent fingerprints are analyzed manually by the experts using the "Analysis, Comparison, Evaluation and Verification (ACE-V)" method [14]. Fig. 5 describes the four phases in the ACE-V method.

The ACE-V examination method is composed of the following phases:

- Analysis (A): In this phase, fingerprint expert conducts initial analysis to determine the suitability of fingerprint for comparison and accordingly assigns one of the three labels: "VID (Value for Individualization), VEO (Value for Exclusion) or NV (No Value)" [15]. If the print is found to be suitable (i.e., the assigned label is VID or VEO), then the fingerprint expert manually marks and indicates all the features.
- Comparison (C): The examiner compares the characteristic features of the latent print (unknown print) indicated in the previous phase with that of the exemplar prints (known prints) and identifies the similarity or dissimilarity between them.
- Evaluation (E): Depending upon the results of the comparison phase, the examiner evaluates and makes a decision in the form of Individualization, Exclusion, or Inconclusive.
- Verification (V): In this phase, another examiner verifies the decision made by the original examiner by independently performing analysis, comparison and evaluation.

Inconsistency in the decision made by the different latent examiners regarding the same latent fingerprint has been studied by Ulery et al. [16]. Hence, need arises for an automated latent fingerprint identification system.

3.2. Latent fingerprint semi-automated analysis

Over the years, remarkable advancements have been made in live-scan fingerprint matching resulting in the large-scale deployment and successful working of automated fingerprint identification systems. However, automated latent fingerprint matching is still a challenging problem as latent fingerprints are generally poor in quality, having less distinctive information. Fig. 6 shows the semi-automated procedure for analyzing latent fingerprints.

First, the latent print examiner develops the latent print using a suitable method (powder, chemical or optical) to make it visible to the naked eye. The developed latent print is lifted or captured. Preprocessing of latent prints is necessary before marking and extraction of features. Therefore, the latent print is segmented from the noisy background and the quality of the ridges is enhanced. Further, features of the latent print lifted from the crime scene are manually marked by the latent print examiner. Usually, all the features (level 1, level 2 and level 3) are examined by the expert. All the marked features are given as input to the Automatic

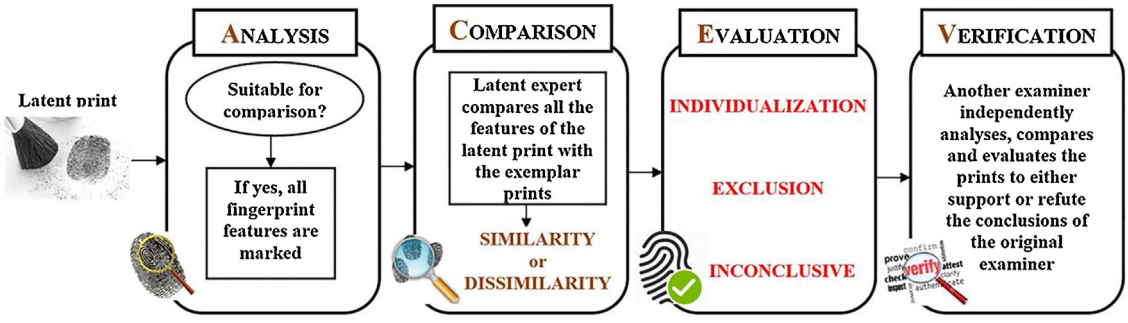


Fig. 5. ACE-V method [14].

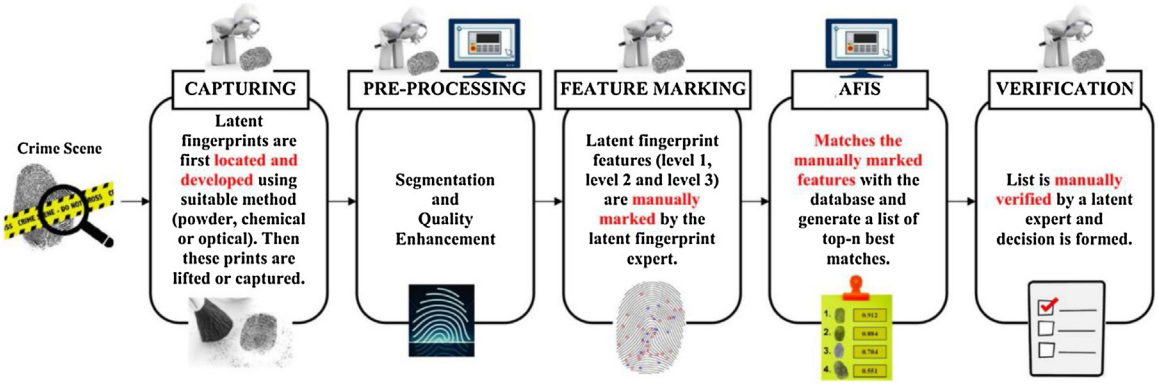


Fig. 6. Semi-automated procedure for analyzing latent prints.

Fingerprint Identification System (AFIS), which matches the features with the database of ten-prints. AFIS system generates a list of best possible matches (top-n; n is usually 50 or 100). The generated list is further verified by the latent expert to determine if the latent print is from the same source.

4. Automated latent fingerprint identification system

Automated Fingerprint Identification System (AFIS) for latent prints involves human intervention for feature marking and verification. A lot of research is going on developing a 'lights-out' latent fingerprint identification system. The process of an automated latent fingerprint identification system can be broadly divided into four sequential processes, namely "segmentation; quality assessment and enhancement; feature extraction and matching" as shown in Fig. 7 [17].

Techniques proposed by eminent researchers for various processes of the automatic latent fingerprint identification system are described below.

4.1. Latent fingerprint segmentation

Latent fingerprints are often found, lifted or captured from a variety of surfaces that differs in texture and pattern. The process of separating the fingerprint portion from the entire captured image is known as segmentation. Accurate segmentation of fingerprint directly influences the process of feature extraction as improper segmentation would result in the extraction of false fingerprint features. It is one of the challenging stages of the AFIS as the foreground fingerprint ridge information is entwined with the background details in most of the cases.

Fingerprints generally possess regional uniformity having a uniform pattern within a small window termed as 'Quasi-global' property of a fingerprint ridge/valley pattern. This property is utilized by Karimi-Ashtiani and Kuo [18] to propose a segmentation technique that is robust to gradient deviation. Local orientation and frequency components of a fingerprint ridge/valley pattern in small windows (blocks) are computed and are reliably assessed using a distance function. Distance value less than the threshold is classified as a fingerprint region, whereas the distance value more than the threshold is classified as a background region.

An approach for segmenting latent fingerprints from a textured or structured background using an ideal ridge template has been proposed by Short et al. [19]. This template is modeled with a sinusoid taking into consideration local contrast, frequency and direction. A pre-processed latent fingerprint is sectioned into local blocks and is cross-correlated with the generated ideal ridge template. A 'goodness of fit' score is computed as a result of cross-correlation and each local block is assigned a quality level that distinguishes foreground and background regions based on the pre-set threshold.

An automated segmentation algorithm for latent fingerprints using frequency and orientation of the ridge has been proposed by Choi et al. [20]. Symmetry features are extracted using orientation tensor, while the local frequency is estimated using the Fourier analysis. The final segmented output is obtained by applying various post-processing steps on the intersection regions of the foreground. The matching performance of the commercial matcher is also improved; however, the algorithm merely performs better for good quality latent fingerprints.

The Total Variation (TV) model is used for image decomposition wherein an image is decomposed into two signal layers, namely cartoon (consisting of a piece-wise smooth component) and texture (consisting of an oscillatory or textured component). As the latent fingerprints are acquired from crime scenes, it contains different types of background noise. TV-L1 model could be used for segmenting the fingerprint from the background containing structured noise patterns. However, some noise patterns and structured noise edge signals have similar scales as that of latent fingerprints. Hence, these are misclassified and extracted as a texture component along with the latent fingerprint. To counter the limitation of the TV-L1 model, Zhang et al. [21] have proposed an adaptive total variation model with spatially variant fidelity for effectively segmenting the latent fingerprint by removing different types of structured background noise and automatically locating the required region. Further, Zhang et al. [22] have extended the work by proposing a Directional Total Variation (DTV) Model considering spatial-dependent texture orientation for latent fingerprint detection and segmentation.

Rúiz-Echartea et al. [23] proposed an image gradient and region-based latent fingerprint segmentation technique. Fingerprint areas have a large gradient magnitude due to abrupt intensity changes as compared to areas having soft intensity changes. Further, region labeling is performed to identify the fingerprint portion using the 8-connectivity of the pixels.

Arshad et al. [24] have used k-means clustering for segmenting the latent fingerprint. Sobel and Tophat filters are used for first enhancing the latent fingerprint. Further, the foreground and background regions are segmented by applying a mask using the clustered data. However, the results of segmenting the ugly latent prints of the NIST SD27 database still need improvement.

A dictionary-based method for latent fingerprint segmentation and enhancement has been proposed by Cao et al. [25]. Image decomposition and coarse to fine estimation using an off-line dictionary are used for segmenting and enhancing the latent fingerprints. Ridge restoration in latent patches is performed using a dictionary of high-quality ridge patches. However, the authors

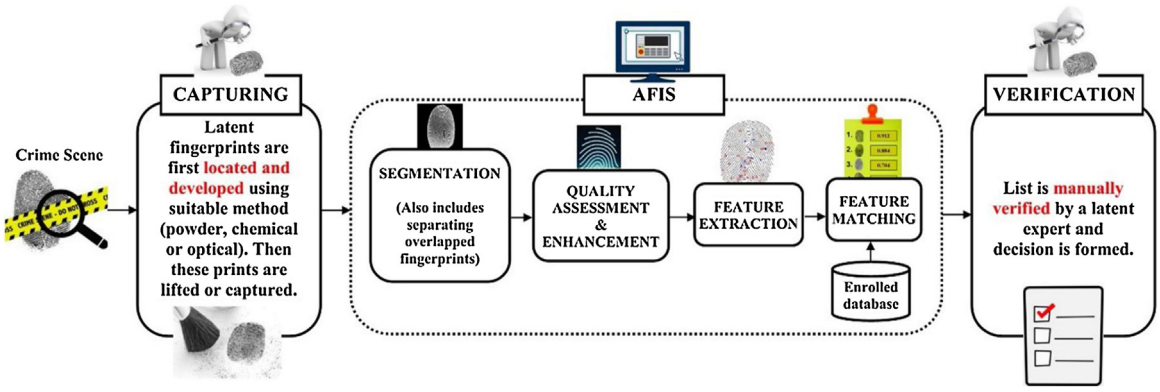


Fig. 7. Processes involved in Automated Latent Fingerprint Identification System [17].

have reported that their approach does not perform well on low-quality fingerprint images and the algorithmic computational efficiency could also be improved.

Revathy et al. [26] have combined fingerprint ridge orientation and local ridge frequency features for segmenting the latent fingerprint. The accuracy of 77% is reported; however, it could be improved by considering additional fingerprint features like ridge types, position and type of minutiae points. Yang et al. [27] have proposed an approach for detecting multiple latent fingerprints using a sequential pose estimation algorithm. Further, detected prints are segmented using a confidence map, which is estimated based on localized dictionaries. Linear density based segmentation algorithm has been proposed by Liu et al. [28]. A Latent image is decomposed into two components, namely texture and cartoon using the TV image model. The texture image is used to detect the linear density map from which a segmentation mask is generated.

In recent years, many machine learning and neural network based segmentation techniques have been proposed by the authors. Stojanovic et al. [29] have proposed a method of segmenting a fingerprint using Convolutional Neural Network (CNN), which is trained with patches of images for classifying the patch as the region of interest or the background. The authors also claimed that the proposed method could be applied to real latent images. Ezeobiejesi and Bhanu [30] have used fractal dimension features for generating feature vectors and Weighted Extreme Learning Machine (WELM) classifier for classifying the regions as fingerprint and non-fingerprint region. Sankaran et al. [31] have proposed a segmentation algorithm using Random Decision Forest. The properties of the fingerprint ridge (foreground) and the noise (background) are determined using five different types of features, namely saliency, image, gradient, ridge and quality based features. Further, an optimum set of features is chosen using a modified Relief algorithm, which is then fed into Random Decision Forest (RDF) classifier to learn and classify the image patches as foreground or background.

Liu et al. [32] proposed a segmentation method making use of the orientation consistency (local feature) and ridge density (global feature) of the fingerprint. Nguyen et al. [33] proposed a method for segmenting the latent fingerprint named SegFinNet. The input image is processed using a fully convolutional neural network and detection based approach. The result shows the improved performance of the state-of-the-art commercial off-the-shelf (COTS) latent fingerprint matcher.

Some of the significant research work done by the researchers in segmenting the latent fingerprint is summarized in Table 2.

Automatic latent fingerprint segmentation is still a challenge. Most of the techniques for segmenting the latent fingerprint perform well for good quality images. However, its performance needs to be improved for poor quality latent prints, which are often found at the crime scenes. Moreover, there are cases wherein two or more latent prints are overlapped. So, separating the overlapped fingerprints becomes even more challenging.

An algorithm for separating overlapped latent prints into individual fingerprints using local Fourier transform has been proposed by Chen et al. [34]. First, the overlapped region in the fingerprint image is manually marked and two candidate orientations are estimated using a local Fourier transform. Then, the relaxation labelling technique is used to label the obtained orientation into two classes, which are further separated into two different orientation fields using the labelling results. Also, the Gabor filter with respect to the individual orientation field is applied to enhance the overlapped fingerprint. However, their proposed approach does not perform well on the fingerprint images containing singular points due to discontinuous orientation fields around these points.

Shi et al. [35] proposed a constraint relaxation labelling algorithm for separating the mixed orientations in the overlapped region. The authors have considered orientations in both overlapped and non-overlapped blocks, whereas in [34], only orientations in overlapped blocks have been considered. However, the authors have reported the scope for the improvement in estimating the initial orientation field. Moreover, the entire separating algorithm is not completely automated as it requires the manual marking of the region masks. Feng et al. [36] further proposed a basic algorithm for estimating the orientation field using constraint relaxation labelling. Also, the authors have proposed the improved version of the basic separation algorithm considering two cases: the first case includes having the knowledge of the mated template of one of the component fingerprints; and another case where two component fingerprints are from the same source.

Relaxation labelling algorithms depend on the orientation field estimation accuracy in the overlapped regions of the fingerprints. However, in case of poor quality overlapped latent prints, it becomes a challenging task to identify the mixed-orientation field accurately. Hence for overcoming this challenge, a model-based separation method for overlapped latent fingerprints is proposed by Zhao and Jain [37] by utilizing the combination model based on the Zero-pole model, Legendre polynomial, 2D Fourier Expansion (FOMFE) and monomial basis function. The proposed model can reconstruct the orientation fields and simultaneously predict unknown orientation using manually marked features.

Zhang et al. [38] proposed an adaptive orientation model fitting based separation algorithm for overlapped fingerprints. In this, authors have used 16 Gabor filters for more accurately estimating the initial orientation fields instead of using local Fourier analysis. Moreover, to separate the orientation fields, the authors have considered both local and global orientation fields. However, it still

Table 2
Techniques used for latent fingerprint segmentation.

Ref.	Year	Technique	Database	Results/Remarks
[18]	2008	Local orientation and frequency components of fingerprint ridge/valley pattern in small windows (blocks).	NIST SD27	Performance is demonstrated visually. Specific metric not available.
[19]	2011	Correlation with ideal ridge template.	NIST SD27	False Acceptance Rate (FAR): 32% True Acceptance Rate (TAR): 68%
[20]	2012	Segmenting the foreground using orientation tensor and local frequency analysis	NIST SD27 WVU DB	Missed Detection Rate (MDR): 14.78% (NIST SD27) 40.88% (WVU DB) False Detection Rate (FDR): 47.99% (NIST SD27) 5.63% (WVU DB)
[21]	2012	Adaptive Total Variation Model	NIST SD27	MDR: 14.10% FDR: 26.13%
[22]	2012	Directional Total Variation Model	NIST SD27	Performance is demonstrated visually. Specific metric not available.
[23]	2012	Image gradient and region detection	FVC2000	Segmentation results are better compared with the segmentation using variance.
[24]	2014	K-means clustering	NIST SD27	MDR: 4.77% FDR: 26.06%
[25]	2014	Dictionary based approach (Course and fine ridge structure dictionary)	NIST SD27 WVU DB	Rank-1 identification accuracy: 61.24% (NIST SD27) 70.16% (WVU DB)
[26]	2014	Combining the segmentation output from orientation and frequency feature of latent prints	NIST SD27	FDR: 47.99% MDR: 14.78%
[27]	2015	Multiple latent prints are detected using sequential pose estimation algorithm and are segmented using confidence map.	NIST SD27 Multi-latent Fingerprint database	Proposed automatic segmentation achieved a comparable performance with manual segmentation.
[28]	2016	Linear density map characterising the interleaved ridge and valley structure.	NIST SD27	MDR: 13.32% FDR: 24.21%
[29]	2016	Deep learning technique (Convolution Neural Network)	FVC2002	Error coefficient: 0.0392 (noisy images) 0.0365 (regular images)
[30]	2016	Fractal Dimension Features and Ensemble Classifier	NIST SD27 WVU DB IIITD	For NIST SD27 MDR: 9.22% FDR: 18.7% For WVU DB MDR: 15.54% FDR: 9.65% For IIITD MDR: 6.38% FDR: 10.07%
[31]	2017	RDF (Random Decision Forest) based latent fingerprint segmentation	NIST SD4 NIST SD27 IIITD-CLF	Rank-50 identification accuracy: 72.3%
[32]	2018	Combining ridge density and orientation consistency	NIST SD27	MDR: 13.03% FDR: 23.17% Average: 18.13%
[33]	2018	FCN based automatic segmentation	NIST SD27 WVU Forensic Database (for training)	For NIST SD27 MDR: 2.57% FDR: 16.36% For WVU MDR: 13.15% FDR: 5.30%

requires the manual marking of the region of interest of component fingerprints. Zhang et al. [39] presented an extended version of [38] and evaluated the proposed algorithm using two different polynomial orientation field models. However, the authors have reported that the proposed approach cannot deal with more than two overlapped fingerprints.

Jeyanthi et al. [40] have proposed an Automatic Recognition System for fingerprints, where overlapped prints are identified using the Adaptive Neuro-Fuzzy Inference System (ANFIS) classifier. The orientation field separation algorithm is used for separating the overlapped fingerprints and the lost orientation fields are reconstructed using the Hough Transform alignment algorithm.

Tejas et al. [41] proposed an automatic approach of region masking for separating overlapped fingerprints. The authors have used basic image processing operations of blurring and thresholding for obtaining overlapped and non-overlapped regions. Stojanovic et al. [42] proposed a neural network based separation technique for overlapped fingerprints using local Fourier analysis and neural networks. Further, the global orientation field enhancement model is used to enhance the orientation field of the component fingerprints.

Table 3 shows the work done by researchers focusing on overlapped latent fingerprints.

Existing research work on separating the overlapped latent fingerprints requires the manual marking of the region of interest. Thus, the completely automated method for separating the overlapped latent fingerprints has not been developed.

4.2. Latent fingerprint quality assessment and enhancement

Quality assessment and enhancement is an essential step before feature extraction. Quality assessment is the process of evaluating the quality of a latent fingerprint by assessing the latent

Table 3
Technique used for separating overlapped latent fingerprints.

Ref.	Year	Technique	Database	Results/Remarks
[34]	2011	Relaxation labeling technique.	Simulated using FVC2002 DB1_B.	Needs improvement if overlapped region contains singular points.
[35]	2011	Constrained Relaxation labeling algorithm	Simulated using FVC2002 DB1 B.	Receiver Operating Characteristic (ROC) Curve indicates the improved performance and fast as compared to [34]
[36]	2012	Basic algorithm based on Constrained Relaxation labeling algorithm	Simulated using FVC2002 DB1 B. Created own database of overlapped fingerprints using powdered method	Needs improvement in separation when overlapped region contains singular points. Manual marking of region masks.
[37]	2012	Modelled fingerprint orientation fields for reconstructing.	Simulated using FVC2002 DB1 B and NIST SD27. Four images of real overlapping latents	Faster as compared to [34,35]. Manual marking of region of interest, singular points and orientation cues
[38]	2014	Adaptive Orientation Model Fitting. Initial orientations are obtained using 16 Gabor filters.	Tsinghua Overlapped Latent Fingerprint Database	Faster as compared to [34]. However, slower as compared to [36]. ROC curve indicates improved performance compared to [35].
[39]	2014	Adaptive Orientation Model Fitting. Initial orientations are obtained using 16 Gabor filters.	Tsinghua Overlapped Latent Fingerprint Database	ROC curve indicates improved performance compared to [36].
[40]	2015	Overlapped fingerprints are identified using Adaptive Neuro Fuzzy Inference System and separated using orientation field.	FVC 2006 DB1-A FVC 2006 DB2-A NIST SD27 SLF	Separation rate: 92.4% Classification rate: 88.6% Accuracy: 92%
[41]	2017	Region of interest is found using basic image processing operations of blurring, erosion and dilation	FVC 2004 FVC 2006	Visually demonstrated the effectiveness of the approach.
[42]	2017	Mixed orientation field in the overlapped region are separated using neural network.	Tsinghua Overlapped Latent Fingerprint Database	Rank 1 Identification Rate for different overlapping ratios: Small – 0.9444 Medium – 0.9138 Large – 0.9091

print after segmentation and ensuring that it is of appropriate quality so that the minimum amount of information required for making a suitable match is available. The quality assessment of the live-scanned rolled or plain fingerprint is usually performed at the sensor level before registering the scanned fingerprint image in the database. Quality assessment for the latent fingerprint is also critical because generally, they are of poor quality and thus may not contain sufficient information for a match. During the quality assessment phase, the examiner assigns a particular latent value label: "VID (Value for Individualization), VEO (Value for Exclusion Only) or NV (No Value)" to an unknown latent print [15]. Large variations in the latent value determination by the examiners have been reported in [43], which arises a need for developing an automated method for accessing the quality of the latent fingerprint.

Yoon et al. [44] proposed a "Latent Fingerprint Image Quality (LFIQ)" measure by defining a latent image quality using level 1 features (average ridge clarity) and level 2 features (minutiae points) for qualitative and quantitative quality respectively. A 2-class classification problem is considered for determining the image quality and further classifying the latent images as VID and non-VID images. Additional features like connectivity of ridges, reliability of minutiae and position of the finger were also considered for determining the latent fingerprint image quality by Yoon et al. [45]. Sankaran et al. [46] studied the effect of local ridge clarity in quality assessment by proposing an automated local ridge clarity and quality assessment algorithm. However, it requires the manually marking of the minutiae points and the Region of Interest (ROI). Chugh et al. [15] studied the fundamentals used by the fingerprint examiner in assigning the latent value by developing a crowdsourcing based framework. Further, a model is trained using crowd-based learning to predict the latent value of fingerprints. Ezeobiejesi and Bhanu [47] proposed a deep learning based latent fingerprint image quality assessment method. Local features from the multiple image patches of the latent image are together used to predict the latent fingerprint quality.

Further, to extract fingerprint features more accurately, noise is removed and the clarity of the ridges of the latent fingerprint is enhanced using various techniques.

Yoon et al. [48] proposed an enhancement algorithm for latent fingerprints. The coarse orientation field is estimated using the skeletonized image generated from commercial Software Development Kit (SDK), which is further regularized and enhanced using a Gabor filter. It requires the manual marking of the ROI and singular points.

Yoon et al. [49] proposed a robust orientation field estimation technique for enhancing the latent fingerprint. First, the orientation element (dominant ridge orientation) in every block is computed using the Short-Time Fourier Transform (STFT). Further, the orientation field is approximated using the Randomized-RAndom SAmple Consensus (R-RANSAC) algorithm and ridge structures are enhanced using Gabor filtering. However, it also requires the manual selection of ROI and marking of singular points.

Dictionary-based latent fingerprint enhancement approaches have been proposed by various researchers. Feng et al. [50] proposed an algorithm for estimating the orientation field by making use of prior ridge structure information. A dictionary of orientation patches obtained from real fingerprints is constructed and further is used to replace a noisy orientation patch with the closest orientation patch from the dictionary. Further, Yang et al. [51] proposed a localized dictionary-based orientation field estimation algorithm. A local dictionary of real orientation patches is maintained and is used to replace the noisy patch at the same location. Kumar and Velusamy [52] pre-processed latent fingerprints by dividing the latent fingerprint into six regions and constructing a region-wise dictionary of overlapped orientation patches. For a latent print, the orientation field of a region is

Table 4
Latent fingerprint enhancement techniques.

Ref.	Year	Technique	Database	Results/Remarks
[48]	2010	Orientation field estimation algorithm using coarse orientation	NIST SD27	Improved matching accuracy for good, bad and ugly latent prints.
[49]	2011	Short-time Fourier Transforms R-RANSAC Gabor filtering	NIST SD27	Performs better than [48] as illustrated by Cumulative Match Curve (CMC) curves.
[50]	2013	Global dictionary of reference orientation patches	NIST SD27	Average estimation error: 18.44 (in degrees)
[51]	2014	Localized dictionaries-based orientation field estimation	NIST SD27	Average estimation error: 14.35 (in degrees)
[52]	2015	Region-wise dictionary of overlapping orientation patches	NIST SD27	Minutiae matching score of 0.8423 for completely overlapping patch
[53]	2017	Gabor and minutiae dictionary	NIST SD27	Recall: 0.4916 Precision: 0.3616
[54]	2015	Estimating orientation field using Convolutional Neural Network	NIST SD27	Improved Rank-1 Identification rate as compared to dictionary based approaches
[56]	2018	FingerNet: CNN based enhancement method.	NIST SD27	Computation speed is less compared to [51].

automatically corrected using lookup function, replacing the corresponding region with the region having the highest similarity index. Xu et al. [53] proposed a two-step multi-scale patch-based sparse representation for ridge enhancement as well as minutiae extraction. Two dictionaries are constructed: gabor and minutiae dictionary. Ridges are enhanced using gabor dictionaries, while both the dictionaries are used for enhancing minutiae.

Cao and Jain [54] posed a classification problem for estimating the orientation field and proposed a ConvNet based approach for the latent fingerprint. Orientation field in a latent patch is identified from the 128 representative orientation patches, which are learned from a large number of the orientation patches. Schuch et al. [55] proposed a regression-based convolution neural network which is trained to estimate the orientation field and reported that regression gives more accurate results as compared to the classification. However, the results are not tested on latent fingerprints. Li et al. [56] present a convolution neural network based latent fingerprint enhancement method named FingerNet. It has three major parts, namely the convolution part for extracting features of the fingerprint, enhancement deconvolution branch for removing noise and orientation deconvolution branch for guiding multi-task learning strategy based enhancement.

Table 4 summarizes the available techniques for enhancing the latent fingerprints.

A literature survey reveals that pre-processing of the latent fingerprint using various enhancement techniques mainly depends on the orientation field. However, estimating the orientation field requires the manual marking of singular points and region of interest, which is considered a limitation in performing automatic enhancement of latent fingerprints. Moreover, there is a need for developing accurate reconstruction algorithms to fill incomplete regions and patches in the latent fingerprints. Also, latent fingerprint enhancement techniques are evaluated based on matching accuracy. However, matching accuracy depends on many more factors. So, specific metrics for evaluating the enhancement need to be developed.

4.3. Latent fingerprint feature extraction

Fingerprint features are mainly categorized into three broad categories [2]. These categories are described below and are shown in Fig. 8.

- Level-1 feature: This includes fingerprint ridge flow (ridge orientation field and pattern type) and singular points (core and delta). These features are generalized and are not unique to each finger. Thus, these features are used for indexing and classifying the fingerprint based on the ridge flow pattern.
- Level-2 feature: This includes distinctive features on the fingerprint ridges known as minutiae points. These are of two types, namely ridge endings and ridge bifurcations. These features can be extracted from the standard 500 ppi images.
- Level-3 feature: These are the dimensional features of the fingerprint ridge, which includes pores, dots, recipient ridges and shape of the ridge. These are highly distinctive features and could be extracted from images having a high resolution (at least 1000 ppi).

The Automatic fingerprint identification system (AFIS) usually makes use of level 1 and level 2 features of the fingerprint for identification. However, features of level 3, which include pores, dots and incipient ridges could be used to establish an identity where there is less number of minutiae points. Thus, this could also help in increasing the identification accuracy of the system.

Jain et al. [57,58] propose an automatic fingerprint matching hierarchical framework by using level 1 (orientation field), level 2 (minutiae points) and level 3 (ridge contours and pores) features. The authors focused on level 3 features as they contribute significant information for discriminating latent fingerprints. Level 1 features are extracted from the template and query fingerprint images. A score is generated after aligning the two images. If the score is more than a pre-defined threshold (i.e., the two images have a similar orientation), then further, level 2 features are extracted. The matching process terminates if the total number of matched minutiae is greater than 12, failing to which the matching process proceeds further to level 3 features. These features are matched locally using the Iterative Closest Point (ICP) algorithm and a match score is generated. However, the authors have not worked on latent images rather have considered plain high-resolution fingerprint images.

Vatsa et al. [59] proposed a quality induced fingerprint identification scheme. The fingerprint quality score is assessed using the Redundant Discrete Wavelet Transform (RDWT). Further, level 3 features are extracted using a level-set curve evolution technique, which are combined with level 1 and level 2 features. Delaunay based indexing algorithm is used for estimating the indexing parameters. Probe and gallery images are matched using these indexing parameters and an indexing score is generated. The indexing score is combined with the quality score and the top 50 best possible matches are used to compute the quality-based likelihood ratio. The authors have considered rolled, slap and latent fingerprints for the experiment. However, all latent fingerprint

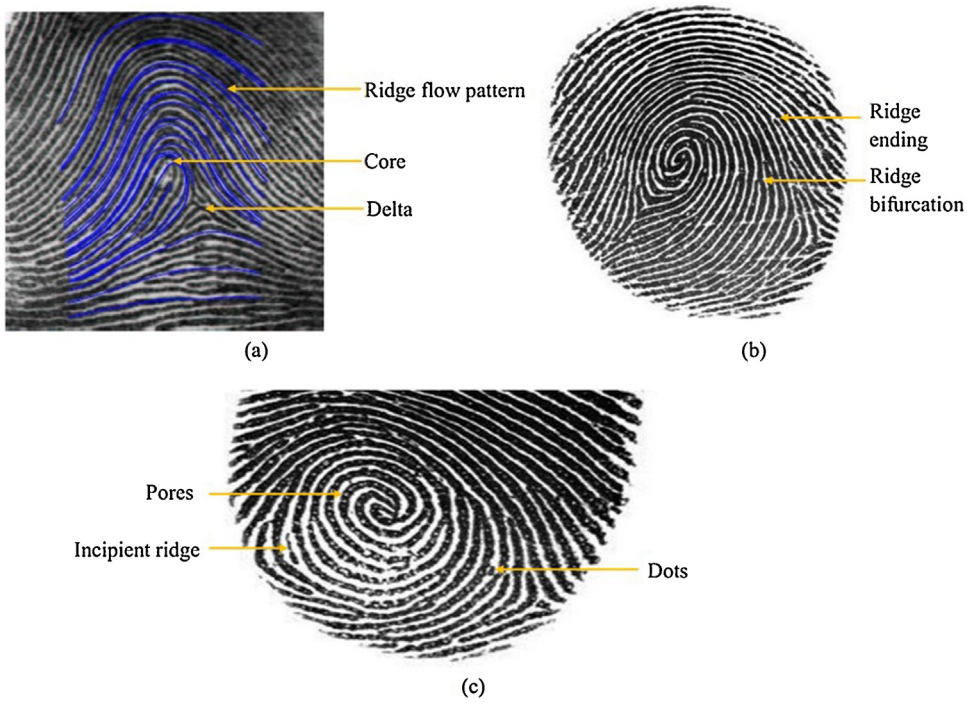


Fig. 8. Fingerprint features (a) Level 1, (b) Level 2, (c) Level 3.

features are manually marked by the professional fingerprint expert.

Zhao et al. [60] propose an adaptive and anisotropic model for extracting the pores based on the observation of the real pores on the fingerprint images. The proposed model can adaptively adjust its parameters using the properties of the local ridges.

Su and Srihari [61] proposed a regression-based gaussian process model to predict the orientation and core point location using the minutiae points. However, the authors reported that the model could not handle latent images of poor quality because of the unavailability of a sufficient number of minutiae points.

Existing pore extraction methods are either skeletonized based, which is computationally expensive and noise-sensitive or filtering-based, which are basically isotropic models having the major limitation of non-adaptability to various fingerprint regions. Hence, an adaptive method for extracting and representing pores more accurately is proposed by Zhao et al. [62]. However, the proposed method is merely for live-scan fingerprints, thus not appropriate for latent fingerprints.

Zhao and Jain [63] have studied the usage and contribution of pores (extended fingerprint feature) to fingerprint recognition accuracy. Pores are extracted using the anisotropic pore model [62] and are matched using a RANdom SAmple Consensus (RANSAC) algorithm [64]. Experiment results show that fingerprint image quality has a great effect on pore extraction as well as matching accuracy. Improvement in matching accuracy using pores is less significant for 500 ppi fingerprint images when compared with 1000 ppi good quality fingerprint images. A fusion of pore with minutiae matcher leads to more accuracy only with good quality fingerprint images. Teixeira and Leite [65] have proposed an approach for extracting pores using spatial and photometric dependence.

Minutiae detection algorithm based on descriptors for latent fingerprints is proposed by Sankaran et al. [66]. Stacked denoising sparse auto-encoders are used for learning descriptors from ten-print fingerprint patches. Further, based on learning, patches are classified as minutia or non-minutiae, considering it as a binary classification problem. However, the proposed method requires manual segmentation of latent fingerprints. Experiment results show that the matching performance in terms of rank-10 identification accuracy is improved as compared with the manual annotation and using commercial matcher (COTS).

Segundo and Lemes [67] have presented an automatic pore-based fingerprint recognition system. Pores are detected using Dynamic Pore Filtering (DPF). Ridges are reconstructed based on Kruskal's minimum spanning tree using consecutive pores considering connect-the-dots strategy. The authors also claimed that the recognition framework is independent of image quality, thus making it suitable for distorted or damaged fingerprints.

Genovese et al. [68] have developed a pore extraction method from the touchless fingerprint images captured using a commercial camera. The candidate sets of sweat pores are extracted using various image processing algorithms. Further, a neural network based technique is used to select the actual sweat pores by learning the local features of the real pores. However, the proposed method is not tested for latent fingerprints.

Labati et al. [69] have proposed a method for extracting pores coordinates from different types of fingerprint images, namely touch-based, touch-less and latent samples using CNN. Results were tested on different databases. For touch-based samples, 30 images from the PolyU HRF database are considered. Dataset for 44 touchless fingerprint images is captured using canon 6D camera at a resolution of 1000 dpi and the dataset for 36 latent fingerprint images is created using the forensic procedure at 1200 dpi. Results show that the method proposed by the authors achieved higher accuracy when compared with the other existing methods.

Su et al. [70] have proposed a pore extraction technique involving CNN to capture the distinctive features of pores. For fingerprint matching, the authors have used the Affline Fourier Moment-matching (AFMM) technique.

Domain knowledge and deep learning methods are combined to form a deep network called FingerNet for efficient and reliable extraction of minutiae by Tang et al. [71]. The proposed FingerNet could be used for extracting minutiae from rolled and latent

fingerprints. However, the performance is highly reliant on the results of the segmentation and enhancement stages. Tang et al. [72] also proposed a fully convolutional network to learn features by adapting to complex background and minutiae are extracted directly without segmentation and enhancement. However, the authors have used harsh thresholds to segment the candidate regions.

Nguyen et al. [73] have presented two network architectures (CoarseNet and FineNet) for automatically extracting minutiae points from the fingerprint image by combining domain knowledge with deep learning. First, a fingerprint image is given as input to the convolution neural network named CoarseNet, which reliably extracts the candidate minutiae locations and orientations by utilizing the domain knowledge. Further, a patch-based classifier named FineNet is used to refine the minutiae locations and orientations by processing each candidate minutiae patch. Finally, precise minutiae locations are obtained by using a non-maximum suppression and thus, boosting the performance of the entire proposed framework.

Table 5 summarizes the various methods used for extracting the features from the latent fingerprint.

While extracting features from a latent fingerprint, spurious minutiae gets generated. Reliable extraction of level 2 features (minutiae points) from poor quality fingerprints continues to be a challenge. Some researchers have focused on the level 3 details, especially pores present on the ridges of the fingerprint, as they contribute significant information for discriminating the latent fingerprints. There are often chances of getting the partial fingerprints at the crime scene wherein only the small part of the fingerprint is available instead of the whole finger impression. Even if the partial print is of good quality, it is highly unlikely to get a sufficient number of level 2 features (minutiae points). So, in

Table 5
Latent fingerprint feature extraction techniques.

Ref.	Year	Technique (Features Used)	Database	Results/Remarks
[58]	2007	Hierarchical recognition framework (Level 1, 2, 3)	Created live-scan database using Crossmatch sensor	Equal Error Rate of the matching system is reduced by 20%
[59]	2008	Level-set curve evolution technique for extracting level 3 features (Level 1, 2, 3)	Created high resolution rolled, slap DB using optical scanner and latent fingerprint database using powder, iodine foaming, ninhydrin and silver nitrate	Rank-20 accuracy using combined level 2, level 3 and quality score for matching rolled with latent: 95.35%
[60]	2008	Adaptive anisotropic pore model (Level 3)	Created live-scan database using own built high-resolution fingerprint scanner	True Detection Rate: 82.8% False Detection Rate: 13.9%
[61]	2010	Regression based gaussian process model (Level 3)	NIST SD27 NIST SD4	Prediction precision: 84.5%
[62]	2010	Dynamic anisotropic pore model and adaptive pore extraction (Level 3)	Created a database of live scan fingerprints using optical fingerprint scanner	True Detection Rate: 84.8% False Detection Rate: 17.6%
[63]	2010	Anisotropic pore model and RANSAC algorithm (Level 2, 3)	NIST SD30 (Rolled ink fingerprint images)	Pore True Detection Rate: 70% False Detection Rate (good): 20% False Detection Rate (medium): 40% False Detection Rate (bad): 70%
[65]	2014	Spatial distribution and photometric dependence (Level 3)	PolyU HRF database	True Rate Detection: 86.1 False Rate Detection: 8.6
[66]	2014	Stacked denoising sparse autoencoders (Level 2)	NIST SD27	Rank 10 identification accuracy: 33.61%
[67]	2015	Dynamic pore filtering (DPF) and Kruskal's minimum spanning tree (Level 3)	PolyU HRF database	Pore: Detection Rate: 90.8% False Detection Rate: 11.1% Ridge reconstruction: Edge Detection Rate: 94.1% False Detection Rate: 3.9%
[68]	2016	Feed Forward Neural Network (FFNN) (Level 3)	Created a touch-less acquired fingerprint database using camera and green LED illumination	Accuracy of FFNN-30: 83.4%
[69]	2017	Convolution Neural Network (CNN) (Level 3)	PolyU HRF database Created a database for latent fingerprints Touch-less database [68]	For Touch-based DB True Detection Rate: 84.69% For Touch-less DB True Detection Rate: 51.8% For Latent DB True Detection Rate: 52.7%
[70]	2017	Convolution Neural Network (CNN) (Level 3)	PolyU HRF database	Pore detection rate: 88.6%
[71]	2017	Deep learning: Convolution Network (Level 2)	NIST SD27 FVC 2004	Precision-Recall curve and CMC curve indicates improvement in minutiae extraction and matching accuracy respectively.
[72]	2017	Fully convolutional network (FCN) (Level 2)	NIST SD 27	Precision: 53.4% Recall: 53%
[73]	2017	Deep Neural Networks (Level 2)	NIST SD27 FVC 2004	Precision: 71.2% Recall: 75.7%

those cases, level 2 information will be insufficient to retrieve the right match from the database. Therefore, in such scenarios, a hybrid feature extraction consisting of level 2 and level 3 features (especially pores) could significantly help in making the individualization decision.

4.4. Latent fingerprint matching

Matching latent fingerprint involves finding the mated match of an unknown latent fingerprint found at the crime scene from the fingerprint database maintained by the law enforcement agency.

Jain et al. [74] proposed a latent fingerprint matching system wherein along with minutiae, some additional fingerprint features are used for matching latent prints such as orientation field and quality map.

Latent fingerprints are matched against the fusion of plain and rolled fingerprints for enhancing the matching performance of latent fingerprints by Fen et al. [75]. It has been found that score level fusion provides a maximum improvement in rank-1 matching accuracy when compared with rank and match level fusion. Dependency on minutiae while matching pore is avoided by proposing a pore-based matching framework by Zhao et al. [64].

Zhao et al. [76] studied the usage of various level 3 fingerprint features and proposed a method using these features for latent fingerprint matching. Apart from considering only the locations of fingerprint features for matching, the authors have also considered the topological relationship between level 2 and level 3 features. The minutiae points from the latent print is matched with the database and the generated rank-1 score is compared against a preset threshold to find out if there is a need for level 3 matching. If the rank-1 score is greater than the threshold, then the list of top N matches is generated as the identification result. Otherwise, level 3 features are compared and the results of minutiae matching is fused with the level 3 feature matching to generate the final identification result.

Manually marked minutiae are combined with the automatically extracted minutiae from the latent fingerprints using score level fusion (boosted-max) by Paulino et al. [77]. First, the orientation field is reconstructed using minutiae and singular points, which are manually marked. Further, the latent print is enhanced and minutiae points are extracted automatically. An enhanced latent print is matched with the rolled print and finally, the scores from the two matchers are fused using the boosted max. However, the authors have reported that orientation field reconstruction needs improvement as inaccurate reconstruction leads to the extraction of spurious minutiae, which eventually affects the matching performance.

Jain and Feng [78] proposed a latent fingerprint matching system using minutiae and extended fingerprint features. Experimental results show that the rank-1 identification accuracy improved to 74% when extended features were used.

Sakaran et al. [79] explored matching two latent fingerprints using different frameworks, namely "decision fusion (OR rule), match score fusion (Product of Likelihood Ratio – PLR) and Support Vector Machine based context switching" algorithm. Three different classifiers, namely NIST Biometric Image Software (NBIS), VeriFinger and Finger-Code have been considered.

Sankaran et al. [80] proposed a hierarchical fusion framework for latent fingerprint matching. First, features are extracted from the fingerprints of the simultaneous impressions and are matched using multiple feature extractors and matching algorithms. Then scores of all the fingerprints in a simultaneously latent impression are fused using weighted sum rule and probability likelihood ratio rule. A list of top k rank is generated since the matching is performed for all the subjects in the gallery for identification. Finally, the rank level fusion is performed using borda count for all top k rank results obtained by multiple feature extractors and matching algorithms.

Yusof and Sulaiman [81] proposed a multiple probes fusion matching framework. Each print in a simultaneous impression is segmented and features are extracted. Multiple probes containing the matching score are created by matching the extracted features with the features extracted from the reference impression of each print. Finally, multi-probe fusion at a score-level is performed.

Paulino et al. [82] proposed a Descriptor-Based Hough Transform approach for matching latent fingerprints. The orientation field reconstructed using manually marked minutiae is aligned and matched with the orientation field of reference print using Descriptor-Based Hough Transform. However, the algorithm for hough transform is complex and time-consuming making it inefficient for AFIS. Zanganeh et al. [83] proposed a region-based fingerprint recognition method in which the correlation coefficient is computed and pixel-wise comparison is made between fingerprints. Zheng and Yang [84] proposed a Minutia Spherical Coordinate Code (MSCC) based latent fingerprint matching algorithm.

Cao and Jain [85] proposed an automated latent fingerprint recognition algorithm wherein a latent is represented by two minutiae templates and one texture template to extract the complementary information from the latent. One minutiae template is generated using ConvNet based ridge-flow, whereas another minutiae template is generated using a ridge flow dictionary. The texture template consists of virtual minutiae. Further, minutiae correspondences are established using second-order and third-order graph-based algorithms. The results of the proposed algorithm are competitive and comparable to the COTS latent matcher.

Nguyen and Jain [86] explored the use of level 3 features (pores) to improve the matching performance of the latent fingerprint identification. The results show that pores improve the matching performance where there is less number of minutiae points.

Table 6 summarizes the work done by the researchers in matching latent fingerprints.

The fair comparison of the performance of the different latent fingerprint matching techniques proposed by the various eminent researchers would not be possible due to the usage of the dissimilar databases. In contrast to plain or rolled fingerprint matching, automated matching of latent fingerprints faces many challenges due to the noisy background, non-linear distortion, partial and fragmented fingerprints, non-availability of singular points and low quality of finger impressions. However, from the literature survey, it could be concluded that the automatic fingerprint identification system for plain and rolled fingerprints has attained sufficient high matching performance while the automated matching of latent fingerprints is still an uphill battle. The next section illustrates the comparison of the different latent fingerprint databases which have been used by the researchers.

5. Latent fingerprint databases

Collection of database for latent fingerprints is a challenging task as it becomes very difficult to simulate the real-time environment of the crime scene having large quality variations and different backgrounds. Latent fingerprint databases being used by various researchers for conducting research are listed in Table 7.

A latent fingerprint database using an optical touch-less device for rigorous testing is not available till now. Also, no public domain latent fingerprint database is available and suitable for studying the utility of level 3 features.

Table 6
Latent fingerprint matching techniques.

Ref.	Year	Technique (features used)	Database	Results/Remarks
[74]	2008	Minutiae and additional fingerprint features (Level 1,2)	NIST SD27	Rank 20 accuracy: 93.4%
[75]	2009	Rank, Match and Score level fusion (Level 2)	ELFT-EFS Public Challenge Dataset	Score level fusion (boosted-max) provided the maximum rank-1 identification accuracy of 83%.
[64]	2009	Refined pores correspondences using RANSAC algorithm and Adaptive anisotropic pore model for pore extraction [60] (Level 3)	Created high resolution live-scan database	30% improvement in recognition Accuracy
[76]	2010	Topological relationship between level 2 and level 3 features (Level 2, 3)	WVU and ELFT-EFS PC Created a simulated partial fingerprint database	Level 3 features can improve the matching accuracy if: it can be reliably extracted from latent as well as plain fingerprints. latent print contains less number of minutiae points.
[77]	2010	Manually marked minutiae are combined with the automatically extracted minutiae using score level fusion (boosted-max). (Level 2)	NIST SD27 NIST SD14	ROI is manually marked for reconstructing the orientation field.
[78]	2011	In addition to minutiae, extended feature set is used for matching latent fingerprints with the plain/rolled fingerprints. (Level 1, 2, 3)	NIST SD27 NIST SD4 NIST SD14	Rank-1 Identification Rate: 74%
[79]	2011	Fusion and context switching frameworks (Level 2)	IIIT-D Latent database Multi latent database	OR rule gives maximum performance compared to PLR (Product of likelihood ratio) fusion. Quality of an image and number of minutiae points are not appropriate parameters for switching among classifiers.
[80]	2012	Combines match score fusion with rank fusion (Level 2)	IIITD SLF database	Simultaneous latent prints can be matched effectively using fusion techniques.
[81]	2012	Multi-probe fusion at score level (Level 2)	IIITD SLF database	Multiple probes improve the matching accuracy of the simultaneous latent print.
[82]	2013	Descriptor-Based Hough Transform (Level 2)	NIST SD27 WVU DB	Rank-1 accuracy of the proposed matcher is better than the VeriFinger.
[83]	2014	Region-Based Similarity Pixel-wise comparison	FVC 2002	Equal Error Rate: 2.02%
[84]	2015	Minutia Spherical Coordinate Code (MSCC) (Level 2)	NIST SD27	Rank-1 Accuracy: 42.2%
[85]	2019	Two minutiae templates and one texture template is utilised to extract the complementary information. (Level 1,2)	NIST SD27 WVU DB	Rank-1 Accuracy: 64.7% (NIST SD27) 75.3% (WVU)
[86]	2019	Minutiae & pore based matcher and re-ranking (Level 2, 3)	NIST SD27 NIST SD30 PolyU-HRF	Rank-1 Accuracy (NIST SD30): 96.4% (Minutiae only) 98.1% (Minutiae and Pores)

Table 7
Latent fingerprint databases.

Name (year)	Size	Resolution	Features extracted	Availability
NIST SD27 Database (2000) [6]	258 latent images along with matching rolled ten-prints	500 ppi	Level 1 Level 2	Not available
IIIT-D Latent Fingerprint Database (2011) [79]	1046 latent images from 15 subjects along with optical slap images	4752 × 3168 Optical sensor images: 500 ppi and 1000 ppi	Level 2	Available
Simultaneous Latent Fingerprint (SLF) Database (2012) [80]	1080 latent images from 30 subjects along with 60 optical slap images	500 ppi	Level 1 Level 2	Available
Tsinghua Overlapped Latent Fingerprint Database (2012) [36]	100 latent and 100 simulated overlapped fingerprints	500 ppi	Level 1 Level 2	Available
ELFT-EFS Public Challenge 2 Database (2012) [8]	1100 latent images	1000 ppi and 500 ppi	Level 1 Level 2 Level 3	Not publicly available
WVU Latent Fingerprint Database (2013) [82]	449 latent images along with 449 mated rolled prints	1000 ppi	Level 2 Level 3	Not publicly Available
IIITD Multi-surface Latent Fingerprint Database (IIITD MSLFD) (2015) [87]	551 latent images from 51 subjects along with 1020 optical slap images	3840 × 2748 Optical sensor images: 1600 × 1500 (500 ppi)	Level 2	Available
Multisensor Optical and Latent Fingerprint Database (2015) [88]	4400 latent images 1600 simultaneous latent images	–	Level 2	Available
PolyU High Resolution Fingerprint Database [89]	DBI: Training: 210 Testing: 1480 DBII: 1480 (Not latent database)	1200 dpi	Level 1 Level 2 Level 3	Available

6. Conclusion

A comprehensive review of the previous research conducted by the eminent authors for developing an automated latent fingerprint identification system is presented in this paper. The steps required for preparing the latent fingerprint before providing as input to AFIS along with the various latent fingerprint development methods for porous and non-porous surfaces are described providing the reader the complete insight of the latent fingerprint development process. Furthermore, the paper discusses the manual and the semi-automated process for analyzing latent fingerprints and highlights the need for the development of a completely automated latent fingerprint identification system.

The commercial products for automatic fingerprint identification system are mostly proprietary and very less information is available at the algorithmic level for completely understanding the computation and algorithms used by such systems. The aim of the paper is to review the techniques proposed by various authors for automating the individual processes of latent fingerprint identification. Existing segmentation, separation of overlapped fingerprints and enhancement techniques usually require the manual marking of the region of interest and other level 1 features. Thus, more exploration could be made towards developing automated segmentation and enhancement techniques for latent fingerprints. Further, level 3 features have significant importance in latent fingerprint matching. Therefore, a hybrid latent fingerprint matching framework using level 1, level 2 and level 3 features needs to be developed.

The development of latent fingerprints using different types of powders or various chemical methods may damage the evidence in case if they are not properly treated. As the technology is improving, new touch-less based latent fingerprint development techniques are also emerging for developing latent fingerprints at the crime scene. Thus, there is a need for the development of a latent fingerprint database using touch-less technology, which could help researchers in testing their proposed techniques on fingerprint images acquired through the latest technology being used in forensics.

CRediT authorship contribution statement

Nancy Singla: Conceptualization, Methodology, Writing - original draft, Writing - review & editing, Visualization. **Manvjeet Kaur:** Supervision, Writing - review & editing, Validation, Visualization. **Sanjeev Sofat:** Supervision, Writing - review & editing.

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